Peter–Weyl theorem and its application

Contest

1. General theory of representation
2. Topological group
3. Peter–weyl theorem
4. Fourier analysis on compact group

1.General theory of representation

Group

Def 1.1 (group)

Group is a set equipped with one binary operation with identity and inverse

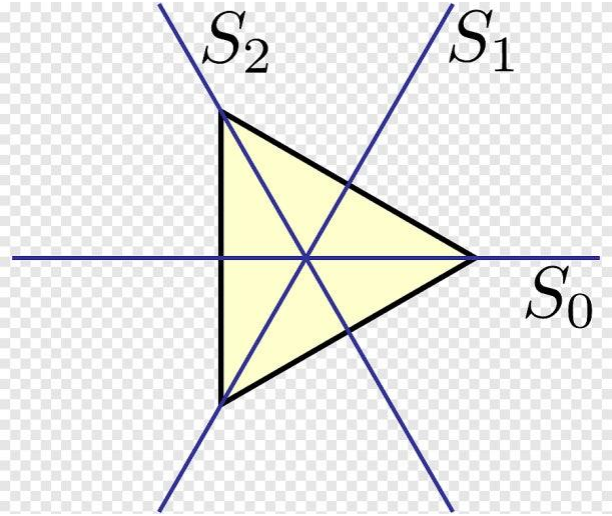


Fig1.1 Dihedral group of triangle

Nature of group as operator

Def 1.2 (group action)

$\cdot : G \times X \to X$ is a group action satisfying followings.

1) $ex = x$

2) $g(hx) = (gh)x$r

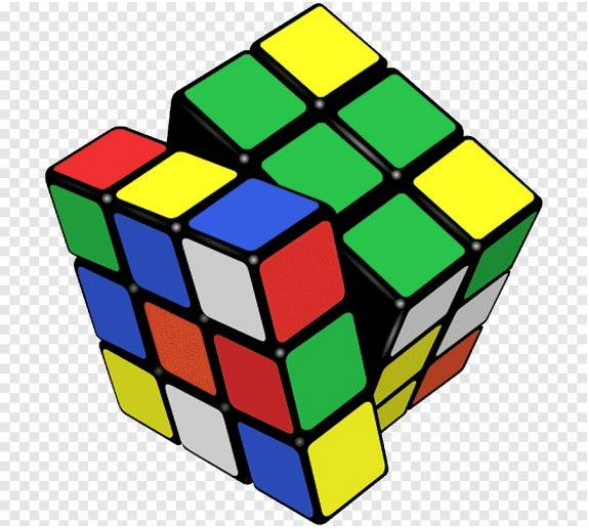


Fig 1.2 Rubik's cube

Prop 1.3

For given an action $\cdot: G \times X \to X$ and for any $g \in G$, $g \cdot: X \to X$ is bijective function. In other words, $\cdot$ is essentially identical with group homomorphism $\rho: G \to Perm(X)$.

Thm 1.4 (Cayley)

For group G, $\rho: G \to Perm(G)$ $\rho(g)h = gh$ is injective homomorphism.

Group representation

Def 1.5 (group representation)

Group representation is a group homomorphism $\rho: G \to GL(V)$ for some vector space V.

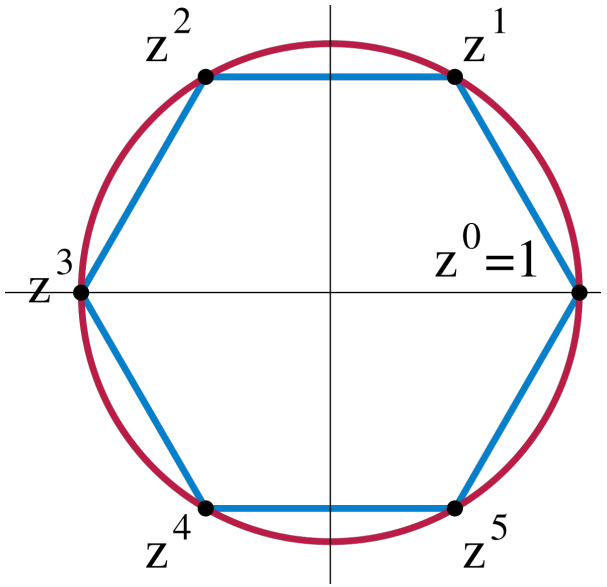
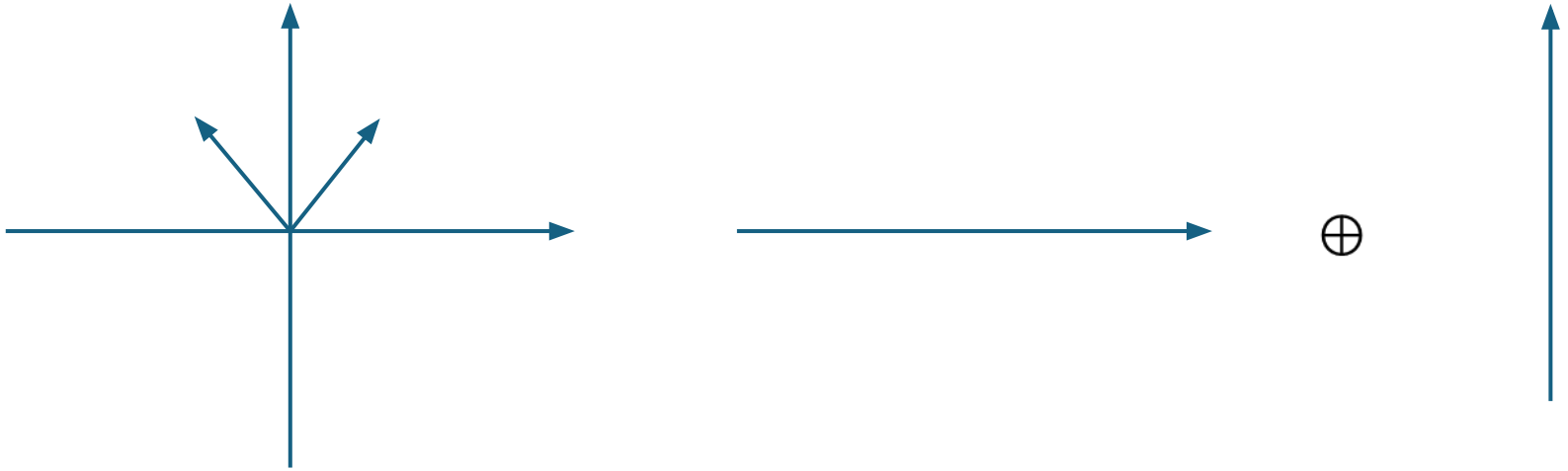


Fig 1.3 Rep of cyclic group

$$\rho(n) = \begin{pmatrix} \cos\frac{n\pi}{3} & \sin\frac{n\pi}{3} \\ -\sin\frac{n\pi}{3} & \cos\frac{n\pi}{3} \end{pmatrix}$$

$Im\ \rho$ → Set of matrix



Equivariant map

- Def 1.6 (equivariant map)

For given two representations of G,

$$\varphi: G \to GL(V) \quad \psi: G \to GL(W)$$

$\alpha: V \to W$ is an equivariant map if following diagram commute.

$$\begin{tikzcd} V \arrow[r, "\alpha"] \arrow[d, "\varphi(g)"] & W \arrow[d, "\psi(g)"] \\ V \arrow[r, "\alpha"] & W \end{tikzcd}$$

Invariant space and irreducible rep

Def 1.7 (Invariant space)

For V equipped with ρ, $W \leq V$ is invariant space if $\rho(g)W \leq W$.

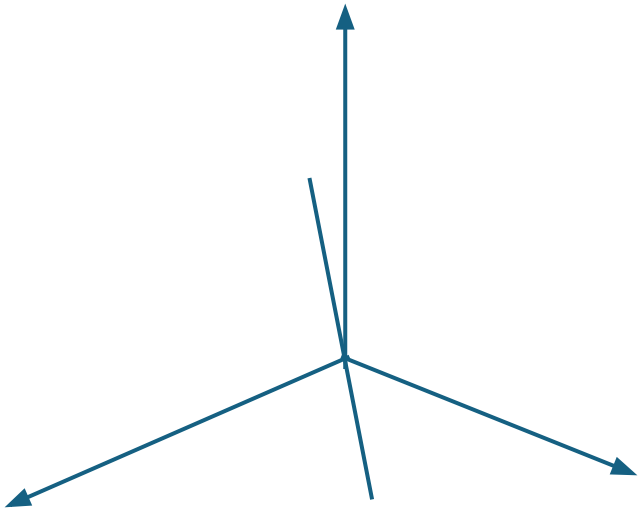


Fig 1.4 Invariant space

Def 1.8 (Irreducible rep)

$\rho: G \to GL(V)$ is irreducible representation if V has no proper invariant subspace.

Example 1.9

-Representation of dimension 1 is irreducible.

-$\rho: Z/2Z \to GL(\mathbb{R}^2)$ $\rho(1) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ is reducible.

$$\rho(n) = \begin{pmatrix} cos\frac{n\pi}{3} & sin\frac{n\pi}{3} \\ -sin\frac{n\pi}{3} & cos\frac{n\pi}{3} \end{pmatrix}$$ is irreducible

Maschke's theorem

Thm 1.10 (Maschke)

If following conditions are satisfied, there exist invariant subspace U s.t. $V = W \oplus U$.

-G is finite

-$char\ F$ does not divide order of G

-W is invariant

Cor 1.11

Every finite dimensional representation of finite group over $\mathbb{C}$ is completely reduced into irreducible representations.

Regular representation

- Lemma 1.12

For finite group G, F^G is F-vector space which has dimension of $|G|$.

Def 1.13 (regular representation)

$r: G \to GL(F^G)$ is left regular representation of G.

$$r_g f(x) = f(g^{-1}x)$$

Thm 1.14

Left regular representation has all irreducible of G as its subrepresentation. Each irreducible representation has multiplicacity of its dimension. i.e. following isomorphism holds.

$$r \simeq \oplus_i \ \oplus_1^{\dim \rho_i} \rho_i$$

$$\begin{matrix} A & 0 & 0 \\ 0 & B & 0 \\ 0 & 0 & \dots \end{matrix}$$

Matrix form

2.Topological group

Topological group

Def 2.1 (topological group)

Group G is topological group if G is topological space and $\cdot : G \times G \to G$ and $^{-1} : G \to G$ are continuous.

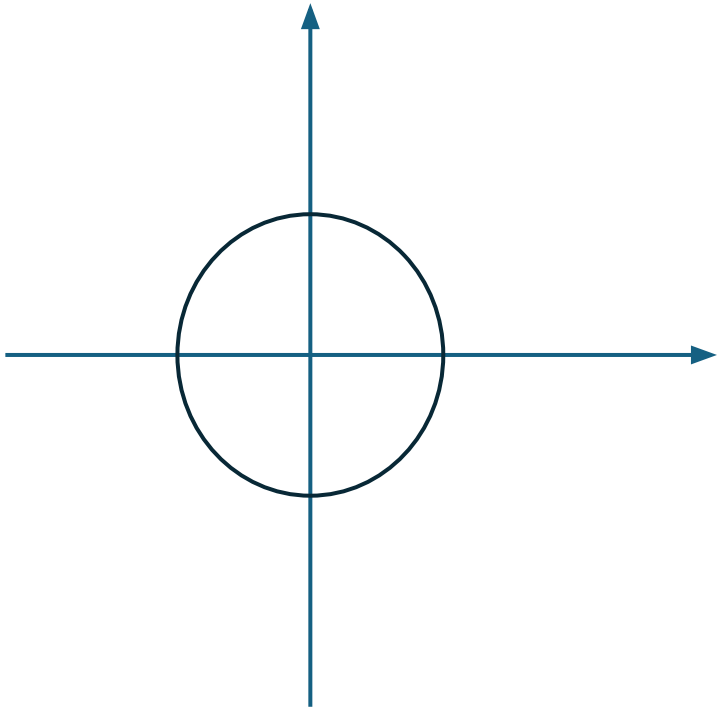
Example 2.2

-Every finite group is topological group with respect to discrete topology

-S^1 is topological group

Def 2.3 (representation of topological group)

For topological group G, $\rho: G \to GL(V)$ is representation, if ρ is continuous homomorphism. Generally, $GL(V)$ has strong operator topology.



$$\rho: S^1 \to GL(\mathbb{R}^2)$$

$$\rho(\theta) = \begin{pmatrix} cosn\theta & sinn\theta \\ -sinn\theta & cosn\theta \end{pmatrix}$$

Haar measure

- Def 2.4 (Haar measure)

For locally compact topological group G, μ is left(right) Haar measure if it satisfy followings.

-μ is inner regular and outer regular

-μ is left(right) invariant. i.e. $\mu(S) = \mu(gS)$

-μ is finite on every compact set

Example 2.5

-Lebesgue measure is Haar measure of $\mathbb{R}^n$

-Counting measure is Haar measure of finite group

Thm 2.6 (uniqueness of Haar measure)

Haar measure is unique up to scaling

3. Peter–Weyl theorem

Decomposition of L2

- Def 3.1 (regular representation)

$r: G \to U(L^2(G))$ $r_g f(x) = f(g^{-1}x)$ is regular representation of compact group G. i.e. translation.

Rmk 3.2

It is not trivial that regular representation of compact group completely decompose into irreducible representation. Generally regular representation has infinite dimension.

- Def 3.3 (convolution)

For $f, g \in L^2(G)$ $(f \star g)(x) = \int f(xh^{-1})g(h)dh$

Prop 3.4

If $\phi \in C(G)$, convolution operator T_ϕ defined by $T_\phi(f) = \phi \star f$ is compact operator. If ϕ satisfy $\phi(g^{-1}) = \overline{\phi(g)}$, T_ϕ is self-adjoint.

Prop 3.5

eigen-space of T_ϕ is invariant space

Peter–weyl theorem

- Thm 3.6 (Peter, Weyl)

  Following isomorphism holds

$$L^2(G) \simeq \bigoplus_{\xi \in \hat{G}} \bigoplus_1^{\dim V_\xi} V_\xi$$

Thm 3.7 (orthonormal basis of L2)

$\mathcal{B} = \left\{ \sqrt{dimV_\xi} < e_i, \rho_\xi(x) e_j > / \, 1 \leq i, j \leq dimV_\xi \right\}$ is orthonormal basis of $L^2(G)$.

4.Fourier analysis on compact group

Fourier series

- Thm 4.1 (trigonometric function forms basis)
  $\{e^{in\theta} \,/\, n \in \mathbb{Z}\}$ forms basis of $L^2(S^1)$

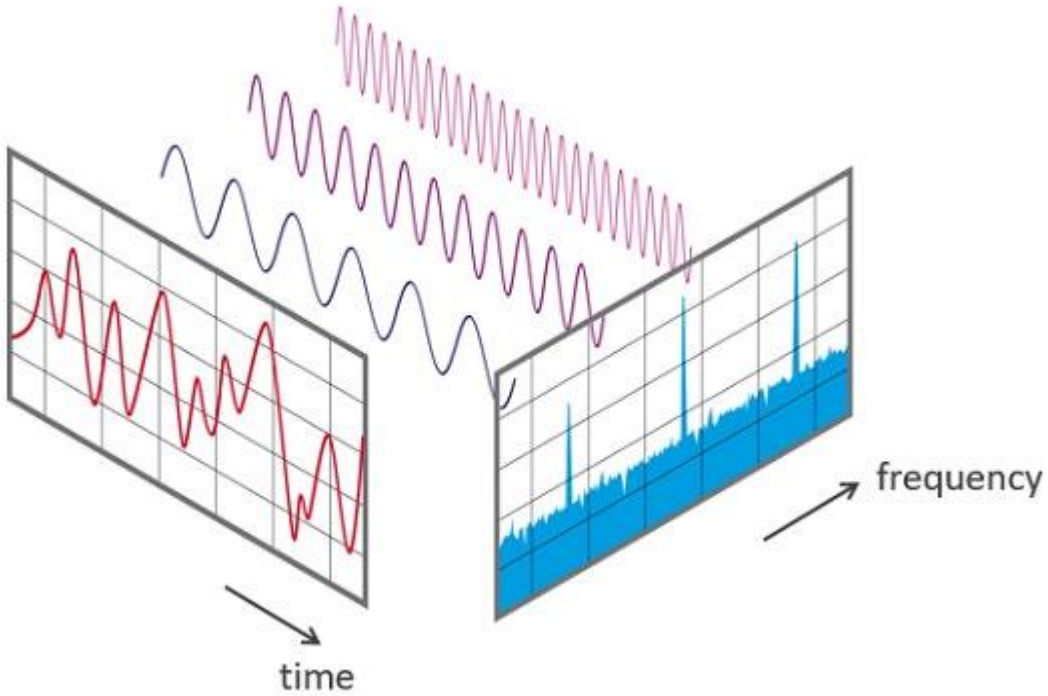


Fig 4.1 Fourier series of S1

- Thm 4.2 (general Fourier series)

$f \in L^2(G)$ can be represented as the Fourier series

$$f = \sum_{\xi \in \hat{G}} \dim(V_\xi) \sum_{i,j=1}^{\dim(V_\xi)} < f, \phi_{ij} > \phi_{ij}$$

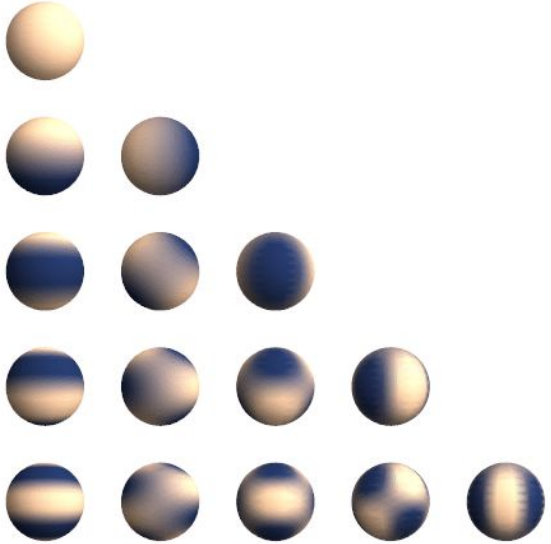


Fig 4.2 spherical harmonics

Plancheral theorem

- Thm 4.4 (Plancherel theorem for S1)

$$\|f\|^2_{L^2(-\pi,\pi)} = \frac{1}{2\pi}\int_{-\pi}^{\pi} |f(x)|^2\, dx = \sum_{n=-\infty}^{\infty} |\hat{f}(n)|^2$$

Thm 4.5 (general Plancherel theorem)

$$\|f\| = \sum_{\xi\in\hat{G}} \dim(V_\xi) \sum_{i,j=1}^{\dim(V_\xi)} \left|< f, \phi_{ij} >\right|^2$$

SU(2)

Def 4.6

Let $P = \mathbb{C}(z, w)$ and P_m is subset of of P consisting of homogeneous polynomial of degree m.

Let $<\cdot,\cdot>$ be inner product of P given by

$$< P, Q > = \int_{S^3} P\bar{Q} d\sigma$$

Prop 4.7

$\{z^m, z^{m-1}w, \dots, zw^{m-1}, w^m\}$ is orthogonal basis of P_m, and $< z^i w^k, z^i w^k > = \frac{j!k!}{(j+k-1)!}$.

There is natural action of SU(2) on P induced by the action of SU(2) on $\mathbb{C}^2$. i.e.

$$U.P(\boldsymbol{z}) = P(U^{-1}\boldsymbol{z})$$

Thm 4.8

Each π_m is irreducible. Moreover, every irreducible representation of SU(2) is isomorphic to π_m for some m.

Prop 4.9

$$< \pi_m e_k, e_j > = \sqrt{\frac{k!(m-k)!}{j!(m-j)!}} \int_0^1 (\bar{a} e^{2\pi i t} - \bar{b})(b e^{2\pi i t + a})^{m-j} e^{-2\pi i k t} dt$$